

Titan Zero New system developers starter template (CRM example)

1) What devs build (3 layers)

A) Data Core (source of truth)

DB tables: tz_* (core ops), ext_* (extensions), no prefix (platform)

Models: app/Models/ (recommended grouping: app/Models/Tz/, app/Models/Ext/)

B) Engines (domain logic)

Engines are business logic services that validate and change state, write events, emit signals.

Put engine logic in: app/Services/Titan<Function>Engine/

Optional domain objects in: app/Domains/<Function>/

C) Systems/Modules (compositions like CRM)

CRM is not a single engine. It's a system that composes: Customer + Talk + Work + Money (and sometimes Docs/Evidence).

Put system logic in: app/Services/Titan<Thing>System/

Put UI/controllers in their normal Laravel places.

2) Canonical folders (use these every time)

Business logic (engines + systems)

app/Services/TitanWorkEngine/*

app/Services/TitanMoneyEngine/*

app/Services/TitanTalkEngine/*

app/Services/TitanSignalEngine/*

app/Services/TitanPulseEngine/*

app/Services/TitanAnalyticsEngine/*

app/Services/TitanCrmSystem/*  (CRM goes here)

Controllers (thin)

app/Http/Controllers/Crm/*

app/Http/Controllers/Work/*

app/Http/Controllers/Analytics/*

Livewire (if used)

app/Livewire/Crm/*

app/Livewire/Work/*

Views

resources/views/panel/crm/*

resources/views/panel/work/*

resources/views/panel/analytics/*

Routes

routes/panel.php → admin/panel UI pages

routes/api.php → PWA/mobile endpoints

Providers (only if you need boot/bindings)

app/Providers/TitanCrmServiceProvider.php (optional)

SQL-first install assets (recommended)

database/sql/ (create this folder)

crm_tables.sql

crm_menus.sql

crm_seed.sql

3) Copy/paste skeleton — TitanCrmSystem

Folder tree

```
app/  
  Services/  
    TitanCrmSystem/  
      CrmService.php  
      CustomerTimelineService.php  
      LeadService.php  
      CrmPermissions.php  
  Http/  
    Controllers/  
      Crm/  
        CrmDashboardController.php  
        CustomersController.php  
  Livewire/  
    Crm/  
      CustomerTable.php      (optional)  
resources/  
  views/  
    panel/  
      crm/  
        index.blade.php  
        customers.blade.php  
  routes/  
    panel.php  
database/
```

```
sql/  
  crm_tables.sql  
  crm_menus.sql
```

4) Copy/paste code — Services

app/Services/TitanCrmSystem/CrmService.php

```
<?php
```

```
namespace App\Services\TitanCrmSystem;
```

```
use App\Models\Tz\TzCustomer;
```

```
class CrmService
```

```
{  
    public function listCustomers(int $teamId, array $filters = [])  
    {  
        $q = TzCustomer::query()->where('team_id', $teamId);  
  
        if (!empty($filters['q'])) {  
            $q->where(function ($qq) use ($filters) {  
                $qq->where('name', 'like', '%'.$filters['q'].'%')  
                    ->orWhere('email', 'like', '%'.$filters['q'].'%')  
                    ->orWhere('phone', 'like', '%'.$filters['q'].'%');  
            });  
        }  
  
        return $q->orderByDesc('id')->paginate(25);  
    }  
  
    public function getCustomer(int $teamId, int $customerId): TzCustomer  
    {  
        return TzCustomer::where('team_id', $teamId)->findOrFail($customerId);  
    }  
}
```

app/Services/TitanCrmSystem/CustomerTimelineService.php

```
<?php
```

```

namespace App\Services\TitanCrmSystem;

class CustomerTimelineService
{
    /**
     * Compose a "CRM timeline" from multiple engines:
     * - customer events
     * - work/job events
     * - invoice/payment events
     * - talk messages
     *
     * Keep it read-only and tenant-scoped.
     */
    public function timeline(int $teamId, int $customerId, int $limit = 50): array
    {
        // Stub: replace with real queries when tables are final.
        return [
            ['type' => 'customer.created', 'at' => now()->toISOString(), 'meta' => ['customer_id' =>
$customerId]],
        ];
    }
}

```

5) Copy/paste code — Controllers (thin)

app/Http/Controllers/Crm/CrmDashboardController.php

```

<?php

namespace App\Http\Controllers\Crm;

use App\Http\Controllers\Controller;

class CrmDashboardController extends Controller
{
    public function index()
    {
        return view('panel.crm.index');
    }
}

```

app/Http/Controllers/Crm/CustomersController.php

<?php

namespace App\Http\Controllers\Crm;

use App\Http\Controllers\Controller;
use App\Services\TitanCrmSystem\CrmService;
use Illuminate\Http\Request;

```
class CustomersController extends Controller
{
    public function index(Request $request, CrmService $crm)
    {
        $teamId = (int) $request->user()->team_id;

        $customers = $crm->listCustomers($teamId, [
            'q' => $request->get('q'),
        ]);

        return view('panel.crm.customers', compact('customers'));
    }
}
```

6) Copy/paste code — Routes

Add to routes/panel.php

```
use App\Http\Controllers\Crm\CrmDashboardController;
use App\Http\Controllers\Crm\CustomersController;

Route::middleware(['auth'])->group(function () {
    Route::prefix('crm')->name('crm.')->group(function () {
        Route::get('/', [CrmDashboardController::class, 'index'])->name('index');
        Route::get('/customers', [CustomersController::class, 'index'])->name('customers.index');
    });
});
```

7) Copy/paste code — Views

resources/views/panel/crm/index.blade.php

```
@extends('default.panel.layout.app')
```

```
@section('content')
```

```
<div class="container">
```

```
<h1>CRM</h1>
```

```
<p>CRM dashboard (signals + customers + conversations + revenue).</p>
```

```
<a href="{{ route('crm.customers.index') }}">Customers</a>
```

```
</div>
```

```
@endsection
```

resources/views/panel/crm/customers.blade.php

```
@extends('default.panel.layout.app')
```

```
@section('content')
```

```
<div class="container">
```

```
<h1>Customers</h1>
```

```
<form method="GET">
```

```
<input type="text" name="q" value="{{ request('q') }}" placeholder="Search customers..." />
```

```
<button type="submit">Search</button>
```

```
</form>
```

```
<div style="margin-top:12px;">
```

```
@foreach($customers as $c)
```

```
<div style="padding:10px;border:1px solid #ddd;margin-bottom:8px;">
```

```
<strong>{{ $c->name }}</strong><br>
```

```
<small>{{ $c->email }} {{ $c->phone }}</small>
```

```
</div>
```

```
@endforeach
```

```
</div>
```

```
{{ $customers->links() }}
```

```
</div>
```

```
@endsection
```

8) Models (recommended pattern)

app/Models/Tz/TzCustomer.php

```
<?php

namespace App\Models\Tz;

use Illuminate\Database\Eloquent\Model;

class TzCustomer extends Model
{
    protected $table = 'tz_customers';

    protected $fillable = [
        'team_id', 'name', 'email', 'phone',
    ];
}
```

9) What devs must NOT do

Don't put business logic in controllers/blades.

Don't write to other engine tables directly (use services/engine APIs).

Don't ignore team_id scoping on any tz_* or ext_* table.